

Implementation of an SOC-based four-stage constant current charger for Li-ion batteries

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ABSTRACT

This study implements a possible use of the state of charge (SOC) instead of the charge voltage limit (Vlimit) to control the charging process for a four-stage constant current charging strategy. To determine the charging current in each stage, an iterative optimization procedure based on Taguchi method is employed to find near-optimal values. To control the change of the charging stage and terminate the charging process, the Coulomb counting method combined with battery's open circuit voltage estimation is adopted for SOC estimation. Tests of Sanyo 840 mAh, 3.6 V lithium-ion (Li-ion) batteries have been conducted with a Keithley 2230-30-1 triple power supply and a Prodigit 3332 F dual electronic load. The implemented charger has an input voltage of 12 V, output currents of 1.176 A, 0.840 A, 0.588 A, and 0.336 A, as well as an output voltage ranged from 0.168 V to 0.588 V. By performing the experiments, the proposed charging strategy has shorter charging time than the equivalent constant current constant voltage (CCCV) and the Vlimit-based charging methods. However, it yields a slightly lower charging efficiency than the equivalent CCCV and pulse current charging methods.

1. Introduction

The standard charging strategy for lithium-ion (Li-ion) batteries consists of constant current-mode (CC) followed by constant voltage-mode (CV), namely the CCCV method. However, this algorithm may not be suitable for fast charging since the CV charging process will significantly extend the charge time [1,2]. Thus, the double-loop control charger and the boost charger were developed to improve the charging performance of the CCCV method [3,4]. Moreover, a combination of time-to-charge (TTC), energy losses (EL), and a temperature rise index as the objective function was proposed to find the analytical solution to the problem of optimally charging a Li-ion battery for the CCCV method with the value of the current in the CC stage being a function of the ratio of weighting on TTC and EL and of the resistance of the battery [5,6].

With regard to its implementation, control techniques, such as fuzzy logic, grey prediction and phase locked loop, have been utilized to optimize the CV-mode charging current [7–10]. Nevertheless, integrating these techniques into a commercial battery charger is not currently available due to requiring more complicated control algorithms.

Alternatively, a multi-stage constant current (MSCC) charging strategy has been shown to have many advantages such as long cycle

life, high charge/discharge energy efficiency and short charging time [11–17]. However, there are two issues raised in conducting this algorithm. One is what is the transition point condition when the charging process switches from one stage to another. This has been solved by setting a charge cutoff voltage (i.e., charge voltage limit, V_{limit}). The other is how to determine the appropriate charging current in each stage, which is regarded as a combinatorial optimization problem. This has been solved by using several algorithms such as fuzzy logic, ant colony systems, Taguchi approach, and consecutive orthogonal arrays [14–19]. Nevertheless, it is difficult to obtain a precise model that determines the optimal charging current value in each stage due to the complex electrochemical properties of Li-ion batteries. Moreover, such a model requires many parameters that are generally difficult to be determined by inexperienced users.

In this study, the state of charge (SOC) is employed as the transition point condition instead of V_{limit} to govern the MSCC charging process. The SOC-based charging method has been demonstrated with four stages and the SOC values at 25%, 50% and 75% are set as the transition points. The experimental results obtained with the proposed charging method are based on the integration of an L_9 orthogonal array (OA) of the Taguchi method (TM) to find a near-optimal charging current pattern, and the Coulomb counting method is combined with

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battery's open circuit voltage (i.e., V_{oc}) estimation in SOC estimation. Possible advantages and disadvantages using SOC as the transition point condition in this study are outlined below [13,19–21]:

- (1) Although the proposed charging method is demonstrated by using four stages with each individual stage of the SOC to be set as 25%, the optimal number of charge stages and the setting range of the SOC in each stage have not yet been proven. Moreover, since the final stage current of the optimal charge current is 0.4 C rate in the demonstrated example, the battery may not have a fully charged capacity with this large current due to the resistance of the battery inside. Nevertheless, it can successfully charge Li-ion batteries with different initial SOC's and has shorter charging time and narrower temperature variation than the equivalent CCCV charging method.
- (2) Unlike evolutionary computation techniques, such as genetic algorithms, particle swarm optimization and ant colony systems that require many parameters that are generally difficult to be determined by inexperienced users, TM offers a scientifically disciplined methodology to explore the search space and select near-optimal values for the parameters.

2. Procedure of searching for the four-stage optimal charging current pattern based on Taguchi's method

2.1. Selecting a proper OA

The first step in TM is to select a proper OA. According to the most commonly used standard OAs for experimental design [14], this study selects the $L_9(3^4)$ OA to perform a search to find the optimal charging current pattern. The $L_9(3^4)$ OA means that only nine experimental runs of searching for the optimal charging current are performed in a 3-level design of charging currents with a 4-stage charging strategy as illustrated in Table 1. With these nine experiments selected on the basis of the TM method, the interactions between the current levels are quasi-uniformly distributed between the columns of Table 1 and are nearly independent [19].

2.2. Assigning each level with a parameter value

To conduct the experiments, the levels in the OA require to be assigned with parameter values. Manufacturers of Sanyo Li-ion battery (UR14500 P) used in this study recommend charging at 0.7 C for the CCCV mode to prolong battery life. However, this study takes a higher charge C-rate with the goal of maintaining the temperature under control during the charge since there is no information about the maximum charge current. To simplify the process of searching for the optimal charging current pattern, the medium level of charging current candidates at Stage 1 (i.e., S_1) in the first iteration is set as two times as much as the nominal charge current for the CCCV mode (i.e., 1.4 C)

Table 1
Illustrative $L_9(3^4)$ OA with the Measured Responses and Their Corresponding SN Ratios.

Exp. no.	Stage name				Measured responses ($j = 1, 2, 3$)	SN ratio
	Stage 1	Stage 2	Stage 3	Stage 4		
1	H	H	H	H	y_{1j}	SN_1
2	H	M	M	M	y_{2j}	SN_2
3	H	L	L	L	y_{3j}	SN_3
4	M	H	M	L	y_{4j}	SN_4
5	M	M	L	H	y_{5j}	SN_5
6	M	L	H	M	y_{6j}	SN_6
7	L	H	L	M	y_{7j}	SN_7
8	L	M	H	L	y_{8j}	SN_8
9	L	L	M	H	y_{9j}	SN_1

Note: H stands for High, M stands for Medium, and L stands for Low.

Table 2

Current Setting Values for the First Iteration.

Stage name	Selected charging current candidates (unit in C rate)		
	H	M	L
Stage 1, S_1	1.5	1.4	1.3
Stage 2, S_2	1.2	1.1	1.0
Stage 3, S_3	0.9	0.8	0.7
Stage 4, S_4	0.6	0.5	0.4

with the difference between a high-level and low-level of 0.1 C as shown in Table 2. In addition, the current between two charging stages at the same level has a standardized difference of 0.3 C that the values at the present stage are smaller than those of previous stages. For instance, the charging current pattern represented by M, L, H and M in experiment no. 6 as tabulated in Table 1 stands for the current sequence of 1.4 C, 1.0 C, 0.9 C and 0.5 C, respectively. In any iteration, the levels of charging currents at each stage are distributed by subtracting a value of 0.05 C. This update is necessary to test a new set of values in the following iterations and therefore cover the full optimization range.

2.3. Applying TM to formulate charging current patterns

In any iteration, the required experiments of a full factorial charging pattern for the four stages at three levels are 81 runs. Based on L_9 OA for the experimental design, nine tests have been performed on 840 mA h Li-ion batteries at a room temperature to search for the optimal charging current pattern. After completing each experiment, such results are considered as the numerical responses of the charging system denoted as y_{ij} . The subscript i is used to reference a particular experiment, while the subscript j is used to reference the battery charging performance where the indices take on values 1, 2 and 3 corresponding to charging efficiency, charging time and temperature variation, respectively.

During the test, the battery's SOC is estimated based on the current integration and V_{oc} estimation. The charging process will move on to the next stage with the charging current at a new preset constant current when the estimated SOC at stages S_1 , S_2 and S_3 is equal to 25%, 50% and 75%, respectively. However, the charging process is terminated as the estimated SOC reaches 100%. After the process of complete charging, the battery is discharged at a constant current of 0.84 A (i.e., 1 C = 0.84 A) until the battery voltage reaches the threshold of 2.6 V. Then, charging efficiency associated with the charging time and temperature variation over the charging process is determined. The charging efficiency of a battery, η , is defined as below:

$$\eta = \frac{I_{\text{discharge}} \times T_{\text{discharge}}}{I_{\text{charge}} \times T_{\text{charge}}} \times 100\% \quad (1)$$

where $I_{\text{discharge}}$ is the discharging current, $T_{\text{discharge}}$ is the discharge duration until a specified cut-off voltage is reached, I_{charge} is the charging current in T_{charge} charging duration at each stage.

The experimental results related to the responses of charging efficiency, charging time and temperature variation in the first iteration of the selected charging currents are shown in Table 3. For example, the experiment no. 8 ($i = 8$) of the second iteration has three responses: charging efficiency ($j = 1$) of 96.38%, charging time ($j = 2$) of 4760 s and temperature variation ($j = 3$) of 2.45 °C. To solve a multi-response optimization problem in the TM, the following steps are suggested and outlined [19]:

Step1: After conducting all the experiments, TM converts the measured responses to the so-called signal-to-noise (SN) ratios, β_{ij} . Since the aim is to maximize the measured responses y_{ij} , the target value of the three responses is classified into two types. One is the larger-the-better (LTB) which is the charging efficiency; the other is the-smaller-the-better (STB) which includes the charging time and temperature

Table 3
Experimental Results for Charging Efficiency, Charging Time and Temperature Variation for the First Iteration.

Exp. no. (i)	Responses					
	Charging eff. for $j = 1$		Charging time (s) for $j = 2$		Temp. var. (°C) for $j = 3$	
	y_{ij} (%)	β_{ij} (dB)	y_{ij} (%)	β_{ij} (dB)	y_{ij} (%)	β_{ij} (dB)
1	95.92	39.638	3850	−71.709	3.12	−9.883
2	96.24	39.667	4386	−72.841	2.75	−8.787
3	95.89	39.635	5035	−74.040	2.21	−6.888
4	96.19	39.663	4767	−73.565	2.22	−6.927
5	95.92	39.638	4246	−72.560	2.49	−7.924
6	95.55	39.605	4342	−72.754	2.38	−7.532
7	95.96	39.642	4528	−73.118	2.62	−8.366
8	96.38	39.680	4760	−73.552	2.45	−7.783
9	95.87	39.634	4217	−72.500	2.40	−7.604

variation. For the two types of responses, the optimal factor level that can maximize the S/N ratio is expressed as follows [19].

$$\beta_{ij} = \begin{cases} -10 \log_{10} \left(\frac{1}{n} \sum_{k=1}^n \frac{1}{(y_{ij,k})^2} \right), & \text{for LTB type response} \\ -10 \log_{10} \left(\frac{1}{n} \sum_{k=1}^n (y_{ij,k})^2 \right), & \text{for STB type response} \end{cases} \quad (2)$$

where n is the number of replicates for $y_{ij,k}$, where $k = 1, 2, 3, \dots, n$. The S/N ratios, β_{ij} , for charging efficiency (i.e., LTB type), charging time and temperature variation (i.e., STB type) as shown in Table 3 are calculated from the first iteration of the experimental results using (2).

Step2: After computing SN_i for every experiment i , the average SN ratio (i.e., $\bar{\beta}_{j,m,n}$) is calculated for the LTB and STB types of responses with four charging stages each having three charging levels ($m =$ charging stages and $n =$ charging levels). The average SN ratios in the first iteration of this example are tabulated in Table 4.

To further illustrate how to obtain the value of $\bar{\beta}_{j,m,n}$, assume that the variable $\bar{\beta}_{2,S_1,H}$ represents the average S/N ratio of the charging time at stage S_1 with a level H charging during the first iteration. Thus, it can be expressed as an average of the S/N ratios for the experiment no. 1, 2

Table 4
Average S/N Ratios for Charging Efficiency, Charging Time and Temperature Variation for the First Iteration (For interpretation of the references to color in this table legend, the reader is referred to the web version of this article).

Responses (unit in dB)	Charging level (n)	Stage number (m)			
		Stage S_1	Stage S_2	Stage S_3	Stage S_4
Charging eff.	H	39.646	39.648	39.641	39.637
	M	39.635	39.662	39.654	39.638
	L	39.652	39.625	39.638	39.659
Charging time	H	-72.863	-72.797	-72.672	-72.256
	M	-72.960	-72.984	-72.969	-72.904
	L	-73.057	-73.098	-73.239	-73.719
Temp. var.	H	-8.519	-8.392	-8.399	-8.470
	M	-7.461	-8.165	-7.773	-8.228
	L	-7.918	-7.341	-7.726	-7.199

Note: The red bold number indicates the largest number in each stage and the level corresponding to that number which has the greatest effects.

and 3 for charging time of the first iteration as follows:

$$\bar{\beta}_{2,S_1,H} = \frac{1}{3}(\beta_{12} + \beta_{22} + \beta_{32}) \\ = \frac{1}{3}[-71.709 + (-72.841) + (-74.040)] = -72.863\text{dB} \quad (3)$$

Similarly, the average S/N ratios of the charging time at stage S_1 with a level M charging and a level L charging during the first iteration can be determined by finding the means of S/N ratios of experiment no. 4, 5, 6 and experiment no. 7, 8, 9, which are $\bar{\beta}_{2,S_1,M} = -72.960\text{dB}$ and $\bar{\beta}_{2,S_1,L} = -73.057\text{dB}$, respectively. Likewise, the values of $\bar{\beta}_{j,S_1,H}$, $\bar{\beta}_{j,S_1,M}$ and $\bar{\beta}_{j,S_1,L}$ for $j = 1$ and 3 can be calculated, and the values of $\bar{\beta}_{j,m,n}$ for stages S_2 , S_3 and S_4 can be calculated from the similar procedure described above. As a result, the average S/N ratios of all the responses for the first iteration can be obtained as shown in Table 4 and the optimal factor levels were colored red and in bold text. For example, the optimal factor levels as indicated in Table 4 are $S_{1,L}$, $S_{2,M}$, $S_{3,M}$, $S_{4,L}$ for charging efficiency, $S_{1,H}$, $S_{2,H}$, $S_{3,H}$, $S_{4,H}$ for charging time, and $S_{1,M}$, $S_{2,L}$, $S_{3,L}$, $S_{4,L}$ for temperature variation. Therefore, the average weight of each factor level is examined to find the balance of these main effects in order to determine the optimal charging pattern.

Step 3: Let $\lambda_{j,m,n}$ be the weight of level n for factor m from response $\bar{\beta}_{j,m,n}$, which is calculated as

$$\lambda_{j,m,n} = \begin{cases} \frac{\bar{\beta}_{j,m,n}}{\max_n(\bar{\beta}_{j,m,n})}, & \text{for LTB type response} \\ \frac{\max_n(\bar{\beta}_{j,m,n})}{\bar{\beta}_{j,m,n}}, & \text{for STB type response} \end{cases} \quad (4)$$

According to (4), the values of $\lambda_{1,S_1,H}$, $\lambda_{2,S_1,H}$ and $\lambda_{3,S_1,H}$ of high level of stage S_1 for the first iteration are calculated for charging efficiency, charging time and temperature variation, and they are 0.9998 ($= \frac{39.646}{39.652}$), 1.0000 ($= \frac{-72.863}{-72.863}$), 0.8758 ($= \frac{-7.461}{-8.519}$), respectively. Likewise, the values of all $\lambda_{j,m,n}$ for stages S_2 , S_3 and S_4 can be determined. To determine the level weight in this study, the values of the weighting factors of charging time, charging efficiency and temperature variation are set to be 3, 2 and 1, respectively. Thus, the average level weight of a high level of stage S_1 for the first iteration is found to be 0.9792 according to (5).

$$\bar{\lambda}_{S_1,H} = \frac{\lambda_{1,S_1,H} \times 2 + \lambda_{2,S_1,H} \times 3 + \lambda_{3,S_1,H} \times 1}{6} \quad (5)$$

Note that the best level for each charging current in each stage is the level with the highest level weight.

Shrinking the Range to Search for the Optimal Charging Current Pattern and Checking the Termination Criterion

As seen from Table 5, the optimal levels of stages S_1 , S_2 , S_3 and S_4 are determined to be M, L, M and L, respectively, since the highest level weights for stages S_1 , S_2 , S_3 and S_4 correspond to $S_{1,M}$, $S_{2,L}$, $S_{3,M}$ and $S_{4,L}$ for the first iteration. Thus, the current sequence of 1.4 C, 1.0 C, 0.8 C and 0.4 C is the optimal charging current pattern for the first iteration. Since the level differences of charging currents of 0.05 C for each stage are chosen as the convergence criteria, the process continues until three iterations are completed. The charging current values for the second and third iterations are tabulated in Table 6. With repeating identical tasks, the values of all $\lambda_{j,m,n}$ and level weights for the second and third iterations can be obtained and are shown in Table 7. Based on the table, the charging current sequence of 1.4 C, 1.0 C, 0.7 C and 0.4 C is found to be the optimal charging current pattern.

3. Evaluation on the SOC estimation

3.1. Identification of Battery model parameters

Fig. 1 shows a battery model used in this study in estimating the dynamic electrical circuit characteristics and nonlinear behaviors of batteries [22,23]. As seen from Fig. 1, the RC circuit on the left is used for SOC tracking and runtime prediction for the battery where the

Table 5

Level Weights for Charging Efficiency, Charging Time and Temperature Variation for the First Iteration (For interpretation of the references to color in this table legend, the reader is referred to the web version of this article).

Responses (unit in dB)	Charging level (n)	Stage number (m)			
		Stage S ₁	Stage S ₂	Stage S ₃	Stage S ₄
Charging eff.	H	0.9998	0.9996	0.9996	0.9994
	M	0.9995	1.0000	1.0000	0.9994
	L	1.0000	0.9990	0.9995	1.0000
Charging time	H	1.0000	1.0000	1.0000	1.0000
	M	0.9986	0.9974	0.9959	0.9911
	L	0.9973	0.9958	0.9922	0.9889
Temp. var.	H	0.8758	0.8747	0.9996	0.9994
	M	1.0000	0.8991	1.0000	0.9994
	L	0.9422	1.0000	0.9995	1.0000
Level weight	H	0.9792	0.9790	0.9865	0.9748
	M	0.9992	0.9819	0.9969	0.9745
	L	0.9890	0.9976	0.9959	0.9944

Note: The red bold number indicates the level selection of charging current for each stage.

Table 6

Selected Charging Current Values for the Second and Third Iterations.

Stage name	Selected charging current values (unit in C rate)					
	Second iteration			Third iteration		
	H	M	L	H	M	L
Stage S ₁	1.45	1.40	1.35	1.45	1.40	1.35
Stage S ₂	1.10	1.00	0.90	1.05	1.00	0.95
Stage S ₃	0.85	0.80	0.75	0.80	0.75	0.70
Stage S ₄	0.50	0.40	0.30	0.45	0.40	0.35

resistor $R_{\text{self-discharge}}$ characterizes the self-discharge energy loss of the battery, the capacitor C_{capacity} as the nominal capacity represents the total stored energy in the battery, the current source i_{cell} denotes the charge or discharge current of the battery, and the voltage across the capacitance V_{soc} varied in the range of the SOC from 0% to 100% quantitatively represents the battery's SOC [23]. The RC circuits simulate the I–V characteristics and transient response of the battery where the voltage-controlled voltage source $V_{\text{oc}}(V_{\text{soc}})$ bridges the non-linear relationship between the SOC (i.e., V_{soc}) to the open-circuit voltage V_{oc} of the battery, the series resistor R_{series} characterizes the charge or discharge energy losses of the battery, two parallel RC networks reflect the short-term and long-term transient responses due to the relaxation effect of battery, and V_{cell} represents the battery's terminal voltage. Note that the self-discharge resistance is ignored in this study as the Li-ion battery has an extremely low self-discharge rate.

According to references [22,23], the equations to estimate SOC for convenience are repeated as below:

$$\text{SOC}(t) = \text{SOC}(0) - (1/C_{\text{capacity}}) \int_0^t i_{\text{cell}}(\tau) d\tau \quad (6)$$

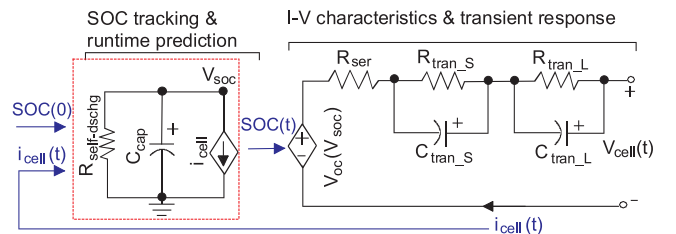
$$V_{\text{cell}}(t) = V_{\text{oc}}(\text{SOC}(t)) - i_{\text{cell}}(t) \cdot R_{\text{series}} - V_{\text{transient}}(t) \quad (7)$$

Table 7

Level Weights for Charging Efficiency, Charging Time and Temperature Variation for the Second and Third Iterations (For interpretation of the references to color in this table legend, the reader is referred to the web version of this article).

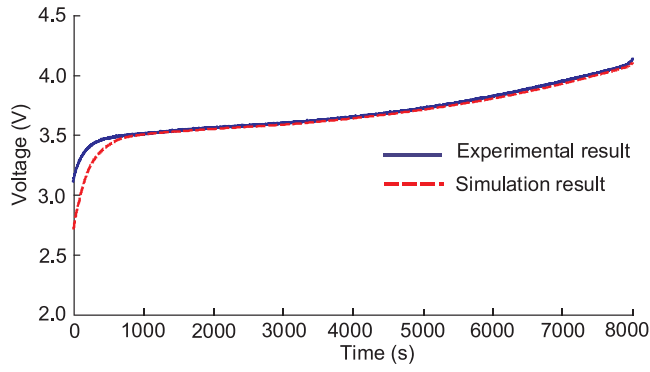
Iter.	Responses (unit in dB)	Charging level	Stage number			
			Stage S ₁	Stage S ₂	Stage S ₃	Stage S ₄
2	Charging eff.	H	0.9993	0.9995	0.9999	0.9988
		M	1.0000	1.0000	1.0000	0.9989
		L	0.9996	0.9992	0.9994	1.0000
	Charging time	H	1.0000	1.0000	1.0000	1.0000
		M	0.9989	0.9978	0.9981	0.9886
		L	0.9984	0.9956	0.9966	0.9724
	Temp. var.	H	0.9107	0.9303	0.9310	0.8756
		M	1.0000	0.9997	0.9565	0.9718
		L	0.9265	1.0000	1.0000	1.0000
	Level weight	H	0.9849	0.9882	0.9885	0.9788
		M	0.9994	0.9988	0.9918	0.9893
		L	0.9868	0.9975	0.9981	0.9862
3	Charging eff.	H	0.9997	1.0000	0.9994	0.9987
		M	0.9998	0.9996	0.9995	0.9986
		L	1.0000	0.9993	1.0000	1.0000
	Charging time	H	1.0000	1.0000	1.0000	1.0000
		M	0.9992	0.9988	0.9981	0.9939
		L	0.9987	0.9978	0.9962	0.9867
	Temp. var.	H	0.9116	0.9178	0.8446	0.8677
		M	1.0000	1.0000	0.8987	1.0000
		L	0.9611	0.9978	1.0000	0.9595
	Level weight	H	0.9851	0.9863	0.9739	0.9775
		M	0.9995	0.9993	0.9820	0.9965
		L	0.9928	0.9983	0.9981	0.9866

Note: The red bold number indicates the level selection of charging current for each stage at the second and third iterations.

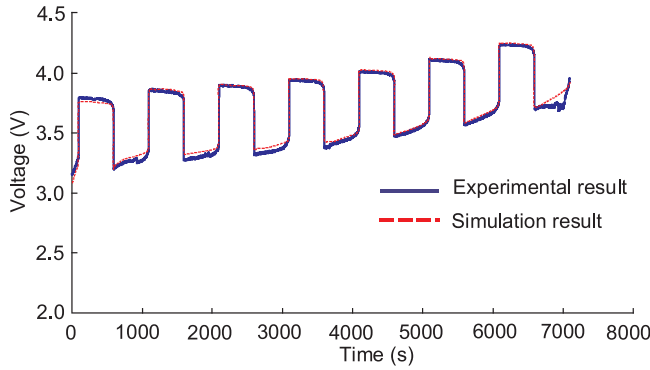
**Fig. 1.** Schematic diagram of the electrical battery model.

$$V_{\text{oc}}(\text{SOC}(t)) = a_0 e^{-a_1 \text{SOC}(t)} + a_2 + a_3 \text{SOC}(t) - a_4 \text{SOC}^2(t) + a_5 \text{SOC}^3(t) \quad (8)$$

where $\text{SOC}(0)$ is the initial SOC, C_{capacity} is the battery nominal capacity, $i_{\text{cell}}(\tau)$ is the instantaneous charge current, and coefficients of $a_1 \sim a_5$ are battery model parameters. The terminal voltage V_{cell} in (7) is estimated by V_{oc} , the voltage across R_{series} , and the transient voltage



(a) With a constant charge current of 400 mA.



(b) With a pulse charge current of 800 mA.

Fig. 2. Comparison between experimental and simulation results of the voltage responses for Sanyo 840 mA h with two charge current operations.

term $V_{\text{transient}}$. As seen from Fig. 1, the RC network parameters are the functions of the SOC and are approximately constant when the SOC is high (e.g., 20–100%) and change exponentially when the SOC varies below a certain value (e.g., 20–0%) due to the electrochemical reaction inside the battery [23].

3.2. Model validation

Some simulation and experiments were carried out to validate the battery model for two charge current operations. Fig. 2 compares the terminal voltage responses obtained from simulations using the battery model with experimental results for a single cell of 840 mA h and 3.6 V for constant-current and pulse-current charge (PCC) scenarios, where the charge currents are 400 mA and 800 mA, respectively. For safety consideration, the battery was charged with a constant current less than an 800 mA and 10-mA end-of-charge current. As seen in Fig. 2, the terminal voltage responses obtained from the simulation using the battery circuit model at 10%–90% SOC agree with those obtained from the experiment. However, the root-mean-square error between them at 0%–10% SOC are about 7.12% and 3.38% for CC and PCC charging scenarios, respectively.

3.3. Relationship between SOC and V_{oc}

To conduct the experiments, equipment including a Keithley 2230-30-1 triple power supply, a Prodigit 3332 F dual electronic load, and a microcontroller board of the arduino mega 2560 interfaced with LabVIEW software program were utilized for further analysis. Li-ion batteries with the nominal capacity of 840 mA h and the nominal voltage of 3.6 V were used in the experiments, and they had a cutoff discharging voltage of 2.6 V and the maximum charging voltage of 4.2 V. To ensure that the battery reaches an equilibrium state at the room

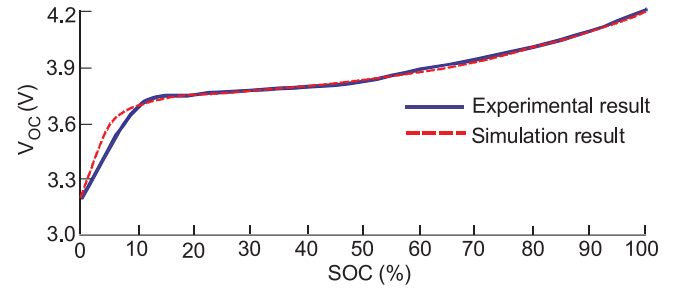


Fig. 3. Comparison of the V_{OC} -SOC curve between simulation and experimental results for the studied Sanyo 840 mA h with a pulse charge current of 0.6 C.

temperature, it was initially fully discharged (SOC = 0%) and rested in the open circuit state for an hour. Then, a selected pulse-charging sequence of 0.6 C consisting of ten minutes charge equivalent to 10% increments of SOC was used to charge the battery and given a ten-minute rest until reaching a fully charged condition (SOC = 100%). Fig. 3 shows a relationship between V_{OC} and SOC based on the experimental and simulation results for the studied battery.

4. Design of the converter and compensator

4.1. Design of the DC/DC buck converter

Assume that all power switches and diodes are ideal elements, and the circuit operates in continuous-conduction mode (CCM). Additionally, an equivalent series resistance r_c associated with the capacitor including the converter equivalent circuit is added to increase the simulation accuracy and a Schottky diode is used to reduce losses. Thus, the minimum values of the inductances, L and capacitances, C in the converter as indicated in Fig. 4 can be obtained using the equations as follows.

$$L > \frac{V_{\text{out}}}{2I_{\text{LB}}} \times (1-D)T_s \quad (9)$$

$$C > \frac{T_s^2 \times V_{\text{out}} \times (1-D)}{8 \times \Delta V \times L} \quad (10)$$

where D is the switch duty cycle, T_s is the switching time, ΔV is the ripple voltage at the load, and I_{LB} is the average inductor currents.

The converter is specified with an input voltage of 12 V, an output voltage of 0.168–0.588 V, an output current of 0.336–1.176 A and a switching frequency of 20 kHz. The minimum values of the inductance and capacitance with selecting the inductor ripple current to be 0.3 times of the output current and the ripple voltage at the load to be 0.05 V were calculated to be 82.17 μH and 42.53 μF , respectively. To ensure the successful operation of the converter in the CCM region and to improve its frequency response of the added compensator network, the selected values of the inductor, capacitor and equivalent series resistance were 120 μH , 360 μF and 75 m Ω , respectively.

4.2. Design of Feedback Compensation Circuit

This study used a Type III compensator to compensate the converter. The crossover frequency was set to 3 kHz. As for the designed converter, the slope of the frequency response of the open-loop control-to-output transfer function of the converter at the first corner ($f_{\text{LC}} = 765 \text{ Hz}$) was -40 dB/dec because the two poles generated from the LC filter fell close to each other. Due to the equivalent series resistance (ESR) associated with a capacitor, a zero occurs a high frequency ($f_{\text{ESR}} = 5894 \text{ Hz}$), resulting in the slope of -20 dB/dec .

To make the closed-loop gain at the crossover frequency to be 0 dB, the simulated gain at the crossover frequency of the open-loop control-to-output transfer function is obtained to be -9.68 dB . Thus, the gain

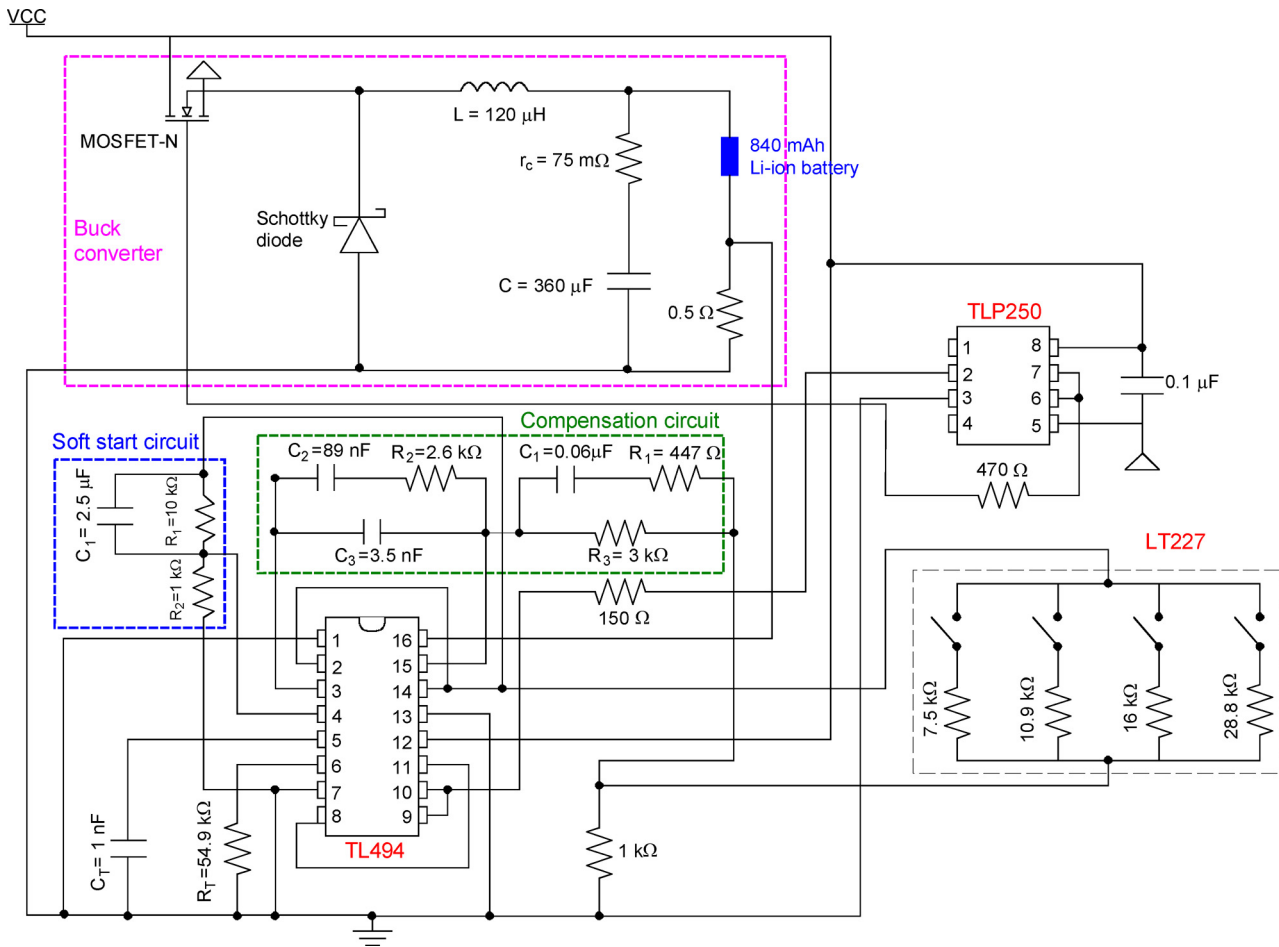


Fig. 4. Basic configuration of the implemented four-stage constant current charging circuit.

of the designed compensation network at the crossover frequency is 9.68 dB. With these conditions, the locations of the two poles are at frequencies of 5894 Hz and 20 kHz, the locations of the zero is at a frequency of 765 Hz, and the corresponding gains are 0.777 and 5.988. Once R_3 is selected as 3 kΩ with the desired poles and zeros, we can find the required values of R_1 , R_2 , C_1 , C_2 , and C_3 to be 447.32 Ω, 2331 Ω, 0.06 μF, 89.25 nF and 3.549 nF. $C_1 = \frac{1}{2\pi \times (R_3 + R_1) \times f_{z1}} = 0.06 \mu\text{F}$. Therefore, the values of electrical elements implemented in the compensation network as shown in Fig. 4 are chosen to be $R_1 = 447 \Omega$, $R_2 = 2.6 \text{ k}\Omega$, $R_3 = 3 \text{ k}\Omega$, $C_1 = 0.06 \mu\text{F}$, $C_2 = 89 \text{ nF}$ and $C_3 = 3.5 \text{ nF}$.

5. Results and discussions

5.1. Testing results on 840 mA h Li-Ion batteries

To verify the performance of the SOC-based FSCC charging strategy, experiments have been carried out with the optimal charging current sequence of 1.176 A, 0.840 A, 0.588 A and 0.336 A (i.e., 1.4 C, 1.0 C, 0.7 C and 0.4 C) that have provided by the output voltages of the designed buck converter of 0.588 V, 0.420 V, 0.294 V, 0.168 V with the current-sense resistance 0.5 Ω, respectively, as shown in Fig. 5. Two Li-ion batteries denoted as B1 and B2 with the same nominal capacity of 840 mA h and the same initial SOC of 0% were chosen to perform the experiments. Fig. 6 shows the experimental results of the charging current pattern, terminal voltage and temperature variation for the batteries. It can be seen that the voltage and temperature profiles are quite consistent for these two batteries.

A comparison is made with CCCV charges with a charging current to

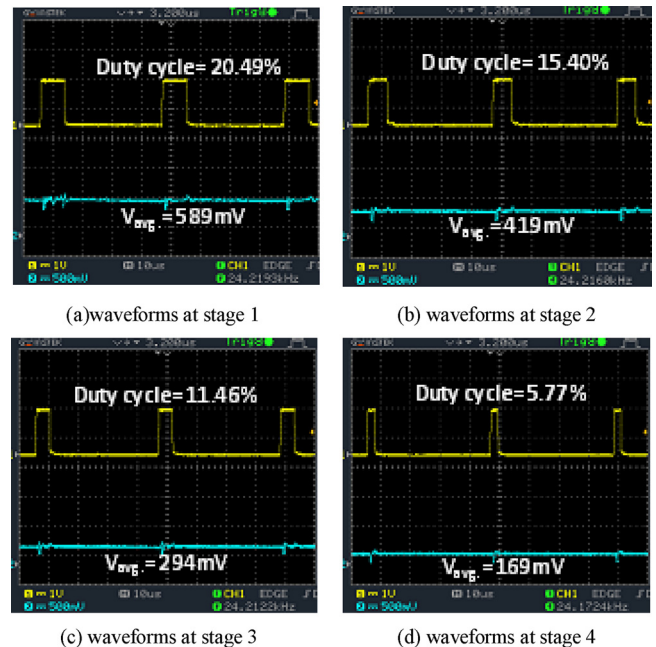


Fig. 5. Waveforms of the constant current charger at each stage. The yellow line is the output waveform of the PWM inverter and the blue line is the output voltage drops across the resistor of 0.5 ohm (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

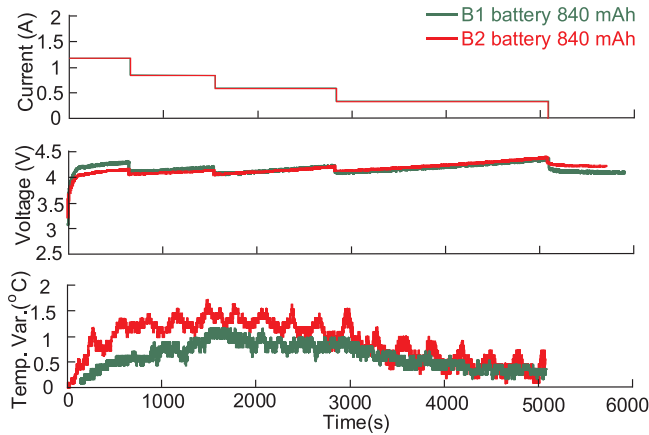


Fig. 6. Experimental results of two different Li-ion batteries using the SOC-based FSCC charging strategy.

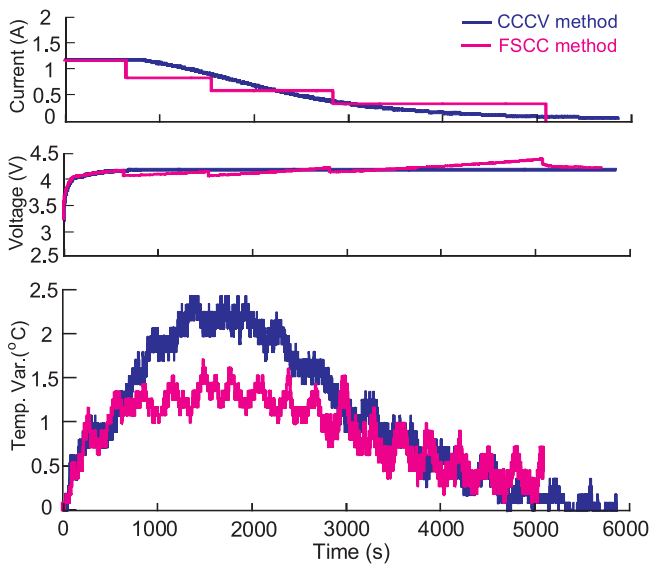
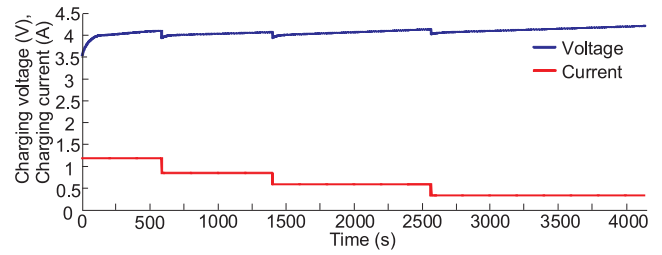


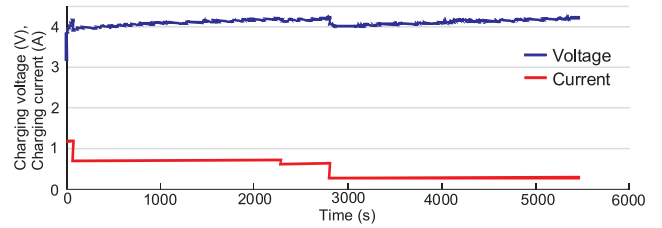
Fig. 7. Comparison between the SOC-based FSCC and CCCV charging methods for the B2 Li-ion battery.

be chosen as the maximum charging current of 1.176 A in the CC stage. In the CV stage, a constant voltage of 4.2 V was used to charge the battery until the battery was fully charged. Fig. 7 shows the experimental results for Li-ion 840 mAh B2 battery that the charging time speeds up when using the FSCC charging strategy. The charge time improves by roughly 15.3% from 98 to 85 min. Moreover, the temperature variation of the SOC-based FSCC charging strategy at stages S_3 and S_4 is smaller than that of the CCCV charging method. Since the theoretical maximum capacity at a low charge current was considered to be 840 mAh, a fully charged battery at 100% SOC will consider to only have 800 mAh instead of 840 mAh according to datasheet [24]. By considering a practical maximum capacity to be 800 mAh, Fig. 8 shows the results with employing the SOC-based FSCC approach compared to other methods. As seen in Fig. 8(c), the battery with the CCCV is charged to

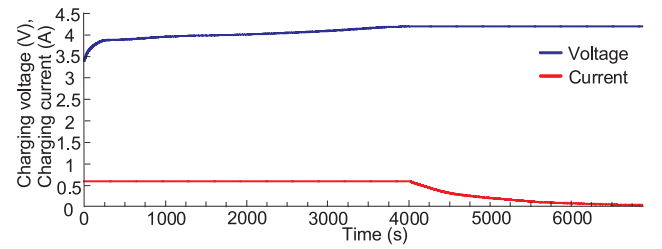
83.0% SOC at time $t = 4019$ s which is at the transition time for switching from CC stage to CV stage. In the SOC-based FSCC method, the corresponding charging SOC is 88.4% as shown in Fig. 8(a). When the battery is charged with only 70% SOC, the charging times based on the CCCV and SOC-based FSCC are 56 min 29 s and 41 min 35 s, respectively. Nevertheless, the voltage at the end of the charging process will drop lower than 4.2 V for both methods as shown in Fig. 9(d) when



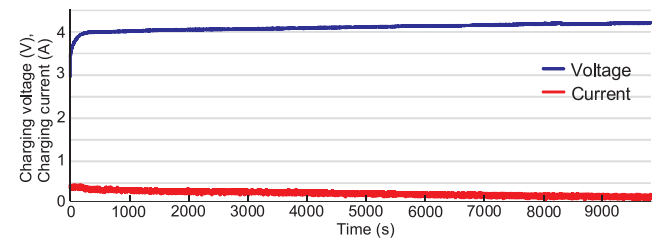
(a) SOC-based FSCC with the charging current sequence of 1.4, 1.0 C, 0.7 C and 0.4 C



(b) V_{lim} -based FSCC with the charging current sequence of 1.40, 0.85 C, 0.75 C and 0.35 C



(c) CCCV with an average charging current of 0.7083 C



(d) CCCV with an average charging current of 0.6040 C

Fig. 8. Experimental results obtained by employing different charging methods for the B2 battery.

the battery is put aside for 30 min. This phenomenon can be attributed to losses during the charging process.

5.2. Testing results on 840 mAh Li-Ion batteries with different initial SOC

Other experiments have been performed to further demonstrate the advantages of employing the SOC-based FSCC charging method. As seen in Fig. 9, the charging processes are terminated at 642 s, 900 s, 1285 s and 2250 s for the corresponding SOC of 25%, 50%, 75% and 100%, respectively. As seen in Fig. 9(d), the maximum charging voltage of the CCCV method is higher than that of the SOC-based FSCC method for the SOC in the ranges of 75–100%. This is because the charge current of the CCCV method (i.e., 0.595 A) is larger than that of the SOC-based FSCC method (i.e., 0.336 A). Although the maximum charge

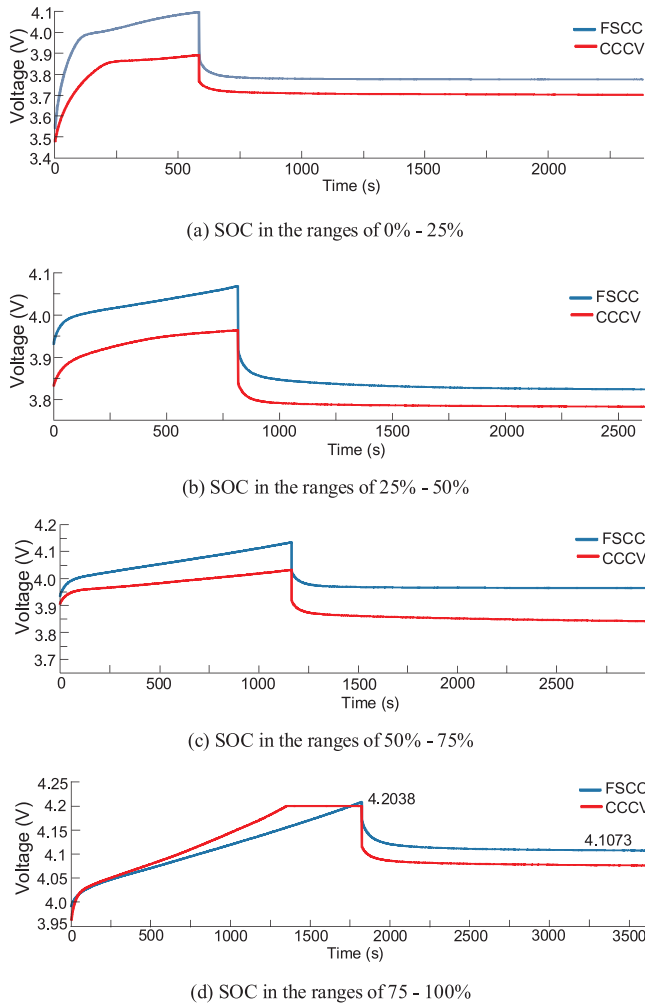


Fig. 9. Experimental results obtained by employing the SOC-based FSCC and CCCV charging methods for the B2 battery with different SOC.

voltage obtained by the SOC-based FSCC method is over the charge voltage limit 4.2 V (i.e., 4.2038 V), it is within the allowed tolerance (i.e., 4.2 ± 0.05 V) [25]. Fig. 10 shows the charging current patterns and the corresponding terminal voltages. It indicates that the SOC-based FSCC charging strategy can successfully charge the Li-ion at any initial SOC (e.g., 17%, 37%, 67% and 84%) which are in the range of 0%–25%, 25%–50%, 50%–75% and 75%–100%) with the assistant of Coulomb counting method combined with battery's open circuit voltage estimation for the SOC estimation which is used to control and terminate the charging process.

5.3. Comparison with the V_{limit} -based FSCC and equivalent CCCV charging methods

Experimental results of using the proposed charging strategy are compared with those of employing the V_{limit} -based FSCC and equivalent CCCV methods to illustrate its performance. The optimal charging current sequence of the V_{limit} -based FSCC charging method with the charge voltage limit of 4.2 V were obtained to be 1.40 C, 0.85 C, 0.75 C and 0.35 C according to the procedure of searching for the four-stage optimal charging current pattern as mentioned in Section 2. In the CCCV charging method, the charging currents of the CC stage were chosen as the average charging currents of the

SOC-based and V_{limit} -based FSCC charging methods and the calculated form is [19]

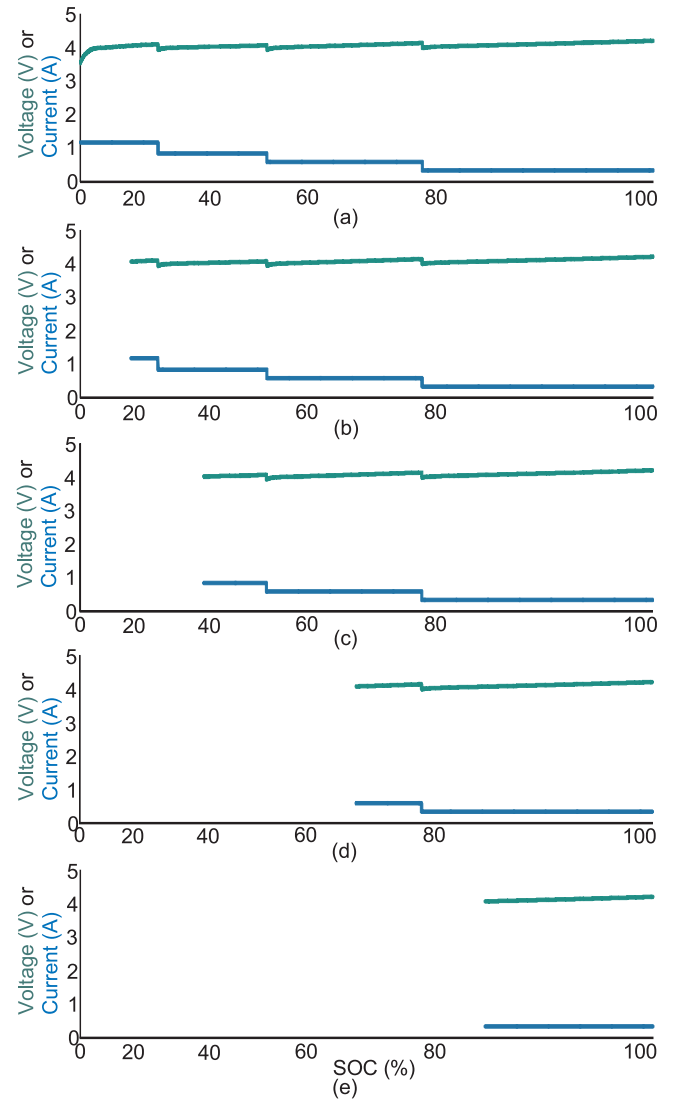


Fig. 10. Experimental results of new charging strategy with different SOC of LiPB B2 (a) SOC = 0%, (b) SOC = 17%, (c) SOC = 37%, (d) SOC = 67%, (e) SOC = 84%.

$$CC_{\text{ave}} = \frac{\sum_{m=1}^4 (I_m \times T_m)}{\sum_{m=1}^4 T_m} \quad (11)$$

where CC_{ave} is the average charging current, I_m is the charging current in T_m charging duration for each stage in the FSCC charging strategy.

The experimental results for the 840 mA h Li-ion battery with different charging algorithms are summarized in Table 8. The charging time of the SOC-based FSCC charging strategy is shorter than that of the V_{limit} -based FSCC charging method (91 min) roughly by 7 min, which is equivalent to 7.88%. The charging time of the SOC-based FSCC charging strategy is shorter than those of the CCCV charging method with the charging currents of 0.7087 C and 0.6040 C roughly by 30 min and 79 min, respectively. In addition, the SOC-based FSCC charging method is compared with the pulse current charging (PCC) strategy where the duty cycle is 50%, the pulse charging frequency is 100 Hz, and the average charging current is 0.595 A (i.e., 0.7083 C) as shown in Fig. 11. As seen in Table 8, the SOC-based FSCC charging method takes roughly 537 more seconds to charge the battery but it's got 18.12% more charged capacity. Table 9 summarizes the differences and advantages of the SOC-based FSCC compared to the prior arts [26]. Note that the charged and discharged capacities as listed in Table 8 are calculated based on equations as follows:

Table 8
Experimental Comparisons of the SOC-based FSCC, V_{limit} -based FSCC, CCCV and PCC Charging Methods.

Charging algorithm	Charging current		Charged capacity (mAh)	Discharged capacity (mAh)	Charging eff (%)	Charging time (s)
SOC-based FSCC	Stage S_1	1.4 C	770.1	742.2	96.4	4520
	Stage S_2	1.0 C				
	Stage S_3	0.7 C				
	Stage S_4	0.4 C				
V_{limit} -based FSCC	Stage S_1	1.40 C	771.4	671.3	87.0	5477
	Stage S_2	0.85 C				
	Stage S_3	0.75 C				
	Stage S_4	0.35 C				
CCCV	1.0 C		600.8	590.3	98.3	9097
	0.7083 C		801.4	777.1	97.0	6871
	(CC_{ave} of the SOC-based FSCC)					
	0.6040 C		688.0	663.3	96.4	9803
PCC	(CC_{ave} of the V_{limit} -based FSCC)					
	0.7083 C		652	642	98.5	3947
(Average charging current of the SOC-based FSCC)						

Note: 1C = 0.84 A, V_{limit} = 4.2 V.

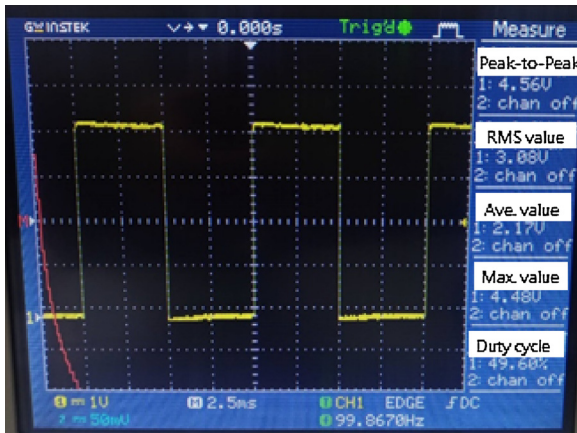


Fig. 11. The waveform of the pulse current charger.

Table 9
The Differences and Advantages of the SOC-based FSCC Compared to the Prior Arts.

Charging protocol	Advantage	Disadvantage
CCCV	Able to be fully charged	Long charging time with reduced life-span
PCC (Pulse current charging)	Long life cycle	Difficult parameters setting
BC (Boost Charging)	Fast charging	High capacity loss and not able to be fully charged
SOC-based FSCC	Fast charging and long life cycle	Not able to be fully charged

$$C_{\text{charge,FSCC}} = \sum_{n=1}^4 \int_0^{t_{\text{cutoff},n}} I_{\text{charge},n} dt \quad (12)$$

$$C_{\text{charge,CCCV}} = \int_0^{t_{\text{cutoff,CCCV}}} I_{\text{charge,CCCV}} dt \quad (13)$$

$$C_{\text{discharge}} = \int_0^{t_{\text{cutoff}}} I_{\text{discharge}} dt \quad (14)$$

where $C_{\text{charge,FSCC}}$ is the FSCC charged capacity, $C_{\text{charge,CCCV}}$ is the CCCV charged capacity, $C_{\text{discharge}}$ is the FSCC or FSCC discharged capacity, $I_{\text{charge},n}$ is the charging current at each stage, $t_{\text{cutoff},n}$ is the time period of 25% SOC, $I_{\text{charge,CCCV}}$ is the CCCV charging current, $t_{\text{cutoff,CCCV}}$ is the time of 0.05 C in the CV stage, V_{cutoff} is the cutoff voltage of 2.6 V, and $I_{\text{discharge}}$ is the discharging current of 1.0 C.

6. Conclusions

This study demonstrates an SOC-based four-stage constant current charging algorithm to speed up the charging process of Li-ion batteries. The proposed method is based on the integration of an L_9 orthogonal array of Taguchi method to find an optimal charging current pattern, and the Coulomb counting method is combined with battery's open circuit voltage estimation for SOC estimation to control the charging process. The performance of the proposed method has been verified by the experimental results of Li-ion batteries and is compared with the performances of the equivalent CCCV charging method, the V_{limit} -based four-stage CC algorithm and PCC charging method. It shows that the SOC-based four-stage CC charging strategy has shorter charging time than the equivalent CCCV and the V_{limit} -based charging methods. However, the proposed charging strategy yields a slightly lower charging efficiency than the equivalent CCCV and PCC charging methods.

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References

- [1] A.A. Hussein, I. Batarseh, A review of charging algorithms for nickel and lithium battery chargers, IEEE Trans. Veh. Technol. 60 (March (no. 3)) (2011) 480–488.
- [2] W. Shen, T.T. Vo, A. Kapoor, Charging algorithms of lithium-ion batteries: an

- overview, Proceedings of IEEE Conference on Ind. Electron. and Appl. Singapore, July 18–20, 2012.
- [3] K.M. Tsang, W.L. Chan, Current sensorless quick charger for lithium-ion batteries, *Energy Convers. Manage.* 52 (March (no. 3)) (2011) 1593–1595.
 - [4] P.H.L. Notten, J.H.G. Op het Veld, J.R.G. van Beek, Boostcharging li-ion batteries: a challenging new charging concept, *J. Power Sources* 145 (July (no. 1)) (2005) 89–94.
 - [5] A. Abdollahi, X. Han, G.V. Avvari, N. Raghunathan, B. Balasingam, K.R. Pattipati, Y. Bar-Shalom, Optimal battery charging, part I: minimizing time-to-charge, energy loss, and temperature rise for OCV-resistance battery model, *J. Power Sources* 30 (January) (2016) 388–398.
 - [6] A. Abdollahi, X. Han, N. Raghunathan, B. Balasingam, K.R. Pattipati, Y. Bar-Shalom, B. Card, Optimal charging for general equivalent electrical battery model, and battery life management, *J. Energy Storage* 9 (February) (2017) 47–58.
 - [7] G.-C. Hsieh, L.-R. Chen, K.-S. Huang, Fuzzy controlled lithium-ion battery charge system with active state of charge controller, *IEEE Trans. Ind. Electron.* 48 (June (no. 3)) (2001) 585–593.
 - [8] L.R. Chen, R.C. Hsu, C.S. Liu, A design of a grey-predicted li-ion battery charge system, *IEEE Trans. Ind. Electron.* 55 (October (no. 10)) (2008) 3692–3701.
 - [9] L.-R. Chen, PLL-based battery charge circuit topology, *IEEE Trans. Ind. Electron.* 51 (June (no. 6)) (2004) 1344–1346.
 - [10] L.R. Chen, J.J. Chen, N.Y. Chu, G.Y. Han, Current-pumped battery charger, *IEEE Trans. Ind. Electron.* 55 (June (no. 6)) (2008) 2482–2488.
 - [11] T. Ikeya, N. Sawada, J. Murakami, K. Kobayashi, M. Hattori, N. Murotani, S. Ujii, K. Kajiyama, H. Nasu, H. Narisoko, Y. Tomaki, K. Adachi, Y. Mita, K. Ishihara, Multi-step constant-current charging method for an electric vehicle nickel/metal hydride battery with high-energy efficiency and long cycle life, *J. Power Sources* 105 (March (no. 1)) (2002) 6–12.
 - [12] T.T. Vo, W. Shen, A. Kapoor, Experimental comparison of charging algorithms for a lithium-ion battery, Proceedings of International Power & Energy Conference, Ho Chi Minh City, Vietnam, DecEmber 12–14, 2012, pp. 1567–1572.
 - [13] A.B. Khan, V.-L. Pham, T.-T. Nguyen, W. Choi, Multistage constant-current charging method for li-ion batteries, Proceedings of IEEE Conference and Expo on Transportation Electrification Asia-Pacific (ITEC), Busan, South Korea, June 1–4, 2016.
 - [14] Y.-H. Liu, C.-H. Hsieh, Y.-F. Luo, Search for an optimal five-step charging pattern for li-ion batteries using consecutive orthogonal arrays, *IEEE Trans. Energy Convers.* 26 (June (no. 2)) (2011) 654–661.
 - [15] Y.-H. Liu, Y.-F. Luo, Search for an optimal rapid-charging pattern for li-ion batteries using the Taguchi approach, *IEEE Trans. Ind. Electron.* 57 (December (no. 12)) (2010) 3963–3971.
 - [16] Y.-H. Liu, J.-H. Teng, Y.-C. Lin, Search for an optimal rapid charging pattern for lithium-ion batteries using ant colony system algorithm, *IEEE Trans. Ind. Electron.* 52 (no. 5) (2005) 1328–1336.
 - [17] J. Jiang, C. Zhang, J. Wen, W. Zhang, S.M. Sharkh, An optimal charging method for li-ion batteries using a fuzzy-control approach based on polarization properties, *IEEE Trans. Veh. Technol.* 62 (September (no. 7)) (2013) 3000–3009.
 - [18] S.C. Wang, Y.H. Lie, A PSO-based fuzzy-controlled searching for the optimal charge pattern of li-ion batteries, *IEEE Trans. Ind. Electron.* 62 (May (no. 5)) (2015) 2983–2993.
 - [19] T.T. Vo, X. Chen, W. Shen, A. Kapoor, New charging strategy for lithium-ion batteries based on the integration of Taguchi method and state of charge estimation, *J. Power Sources* 273 (January) (2015) 413–422.
 - [20] F. Naznin, State of charge governed fast charging method for lithium based batteries, Proceedings in COMSOL Conference, October 17–18, Bangalore, India, 2013.
 - [21] X. Zhang, W. Zhang, H. Li, Review of state of charge estimation methods for li-ion batteries, *Trans. Electr. Electron. Mater.* 18 (June (no. 3)) (2017) 136–140.
 - [22] T. Kim, W. Qiao, A hybrid battery model capable of capturing dynamic circuit characteristics and nonlinear capacity effects, *IEEE Trans. Energy Convers.* 26 (December (no. 4)) (2011) 1172–1180.
 - [23] M. Chen, G.A. Rincón-Mora, Accurate electrical battery model capable of predicting runtime and I–V performance, *IEEE Trans. Energy Convers.* 21 (June (no. 2)) (2006) 504–511.
 - [24] W.-Y. Chang, The state of charge estimating methods for battery: a review, ISRN Applied Mathematics, Hindawi Publishing Corporation, 2013.
 - [25] (2018, April), [Online]. Available: http://batteryuniversity.com/learn/article/charging_lithium_ion_batteries.
 - [26] E. Ayoub, N. Karami, Review on the charging techniques of a li-ion battery, Third International Conference on Technological Advances in Electrical, Electronics and Computer Engineering (TAECE), Beirut, Lebanon, April 29–May 1, 2015.
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